MLOPS ASSIGNMENT

Assignment 1 – Apache Pig

Student Name	Dongari Sai Pallavi
USN	4GW23CI046
Class / Section	7th Sem, CSE(AI&ML)

1. Problem Statement

An e-commerce company wants to prune invalid product reviews and find the average rating score per product category using procedural dataflow scripts. The task is to write a Pig Latin script that loads the review dataset from HDFS, removes invalid ratings, groups valid reviews by product category, and computes the average rating for each category.

2. Tools and Technologies Used

- Ubuntu Linux
- Java 17
- Apache Hadoop 3.3.6
- HDFS
- YARN / MapReduce
- Apache Pig 0.17.0
- Pig Latin

3. Dataset

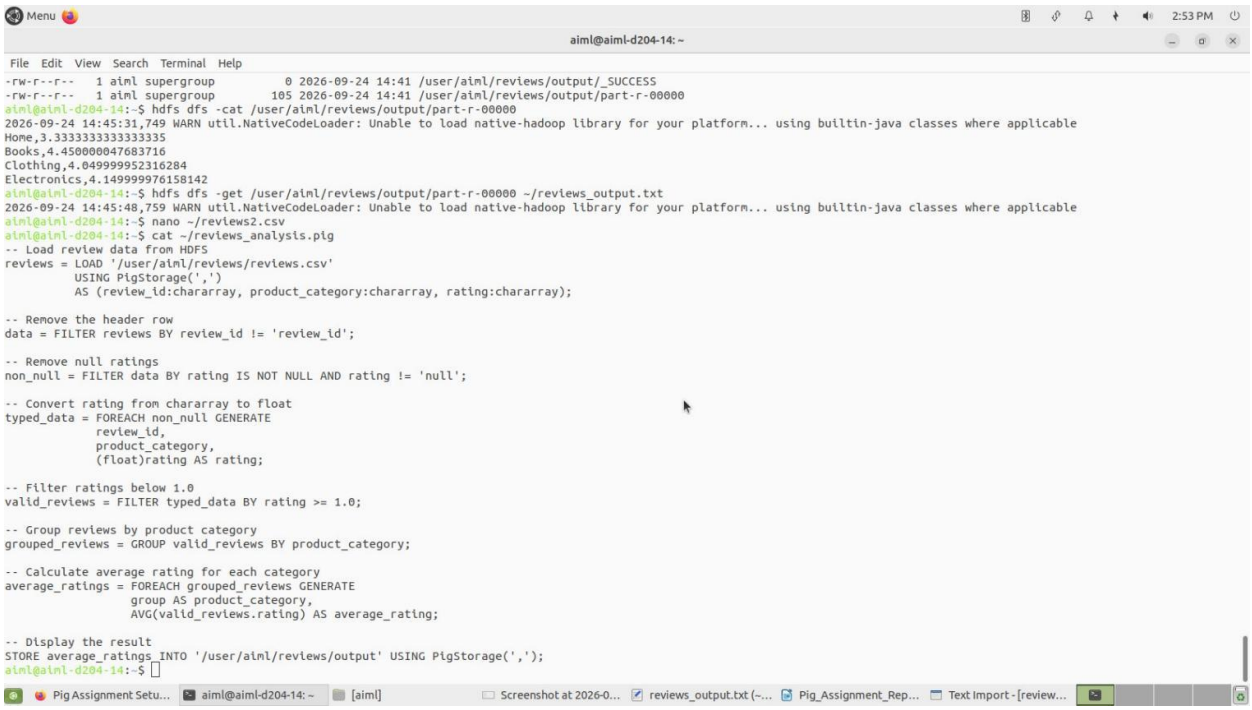
The review dataset is stored as a CSV file named reviews.csv and uploaded to HDFS under /user/aiml/reviews/reviews.csv.

Dataset fields: Field	Type used in Pig	Description
review_id	chararray	Unique review identifier
product_category	chararray	Product category
rating	chararray → float	Review rating score

Dataset contents:

```
review_id,product_category,rating
R001,Electronics,4.5 R002,Electronics,3.8 R003,Electronics,0.5 R004,Clothing,4.2
R005,Clothing,3.9 R006,Clothing,null R007,Books,4.8 R008,Books,4.1 R009,Books,0.0
R010,Home,3.5 R011,Home,4.0 R012,Home,2.5
```

4. Code Implementation

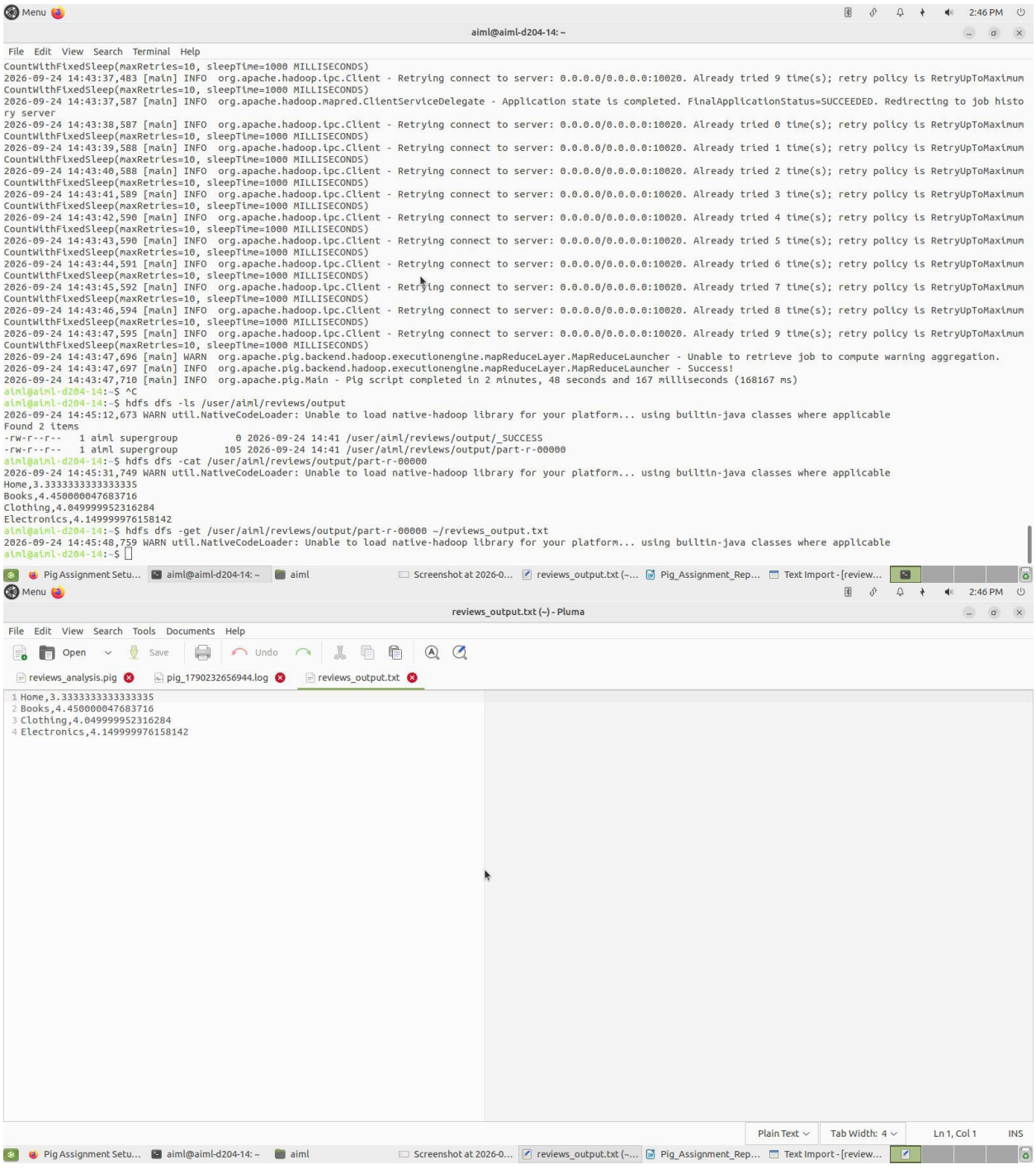


5. Execution

The Pig program was executed in MapReduce mode using:

```
pig -x mapreduce ~/reviews_analysis.pig
```

The program was successfully submitted to YARN and completed the MapReduce processing. The final terminal output is shown below.


Menu
2:46 PM
aiml@aiml-d204-14: ~
File Edit View Search Terminal Help
CountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-24 14:43:37,483 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 9 time(s); retry policy is RetryUpToMaximum
CountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-24 14:43:37,587 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-24 14:43:38,587 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 0 time(s); retry policy is RetryUpToMaximum
CountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-24 14:43:39,588 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 1 time(s); retry policy is RetryUpToMaximum
CountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-24 14:43:40,588 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 2 time(s); retry policy is RetryUpToMaximum
CountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-24 14:43:41,589 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 3 time(s); retry policy is RetryUpToMaximum
CountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-24 14:43:42,590 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 4 time(s); retry policy is RetryUpToMaximum
CountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-24 14:43:43,590 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 5 time(s); retry policy is RetryUpToMaximum
CountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-24 14:43:44,591 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 6 time(s); retry policy is RetryUpToMaximum
CountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-24 14:43:45,592 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 7 time(s); retry policy is RetryUpToMaximum
CountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-24 14:43:46,594 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 8 time(s); retry policy is RetryUpToMaximum
CountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-24 14:43:47,595 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 9 time(s); retry policy is RetryUpToMaximum
CountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-24 14:43:47,696 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Unable to retrieve job to compute warning aggregation.
2026-09-24 14:43:47,697 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-24 14:43:47,710 [main] INFO org.apache.pig.Main - Pig script completed in 2 minutes, 48 seconds and 167 milliseconds (168167 ms)
aiml@aiml-d204-14:~$ ^C
aiml@aiml-d204-14:~$ hdfs dfs -ls /user/aiml/reviews/output
2026-09-24 14:45:12,673 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 2 items
-rw-r--r-- 1 aiml supergroup 0 2026-09-24 14:41 /user/aiml/reviews/output/_SUCCESS
-rw-r--r-- 1 aiml supergroup 105 2026-09-24 14:41 /user/aiml/reviews/output/part-r-00000
aiml@aiml-d204-14:~$ hdfs dfs -cat /user/aiml/reviews/output/part-r-00000
2026-09-24 14:45:31,749 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Home,3.3333333333333335
Books,4.450000047683716
Clothing,4.049999952316284
Electronics,4.149999976158142
aiml@aiml-d204-14:~$ hdfs dfs -get /user/aiml/reviews/output/part-r-00000 ~/reviews_output.txt
2026-09-24 14:45:48,759 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
aiml@aiml-d204-14:~$
Pig Assignment Setu...
aiml@aiml-d204-14: ~
aiml
Screenshot at 2026-0...
reviews_output.txt (~...
Pig_Assignment_Rep...
Text Import - [review...
Menu
2:46 PM
reviews_output.txt (~) - Pluma
File Edit View Search Tools Documents Help
Open
Save
Undo
reviews_analysis.pig
pig_1790232656944.log
reviews_output.txt
1 Home,3.3333333333333335
2 Books,4.450000047683716
3 Clothing,4.049999952316284
4 Electronics,4.149999976158142
Plain Text
Tab Width: 4
Ln 1, Col 1
INS
Pig Assignment Setu...
aiml@aiml-d204-14: ~
aiml
Screenshot at 2026-0...
reviews_output.txt (~...
Pig_Assignment_Rep...
Text Import - [review...

6. Dataflow

The implemented procedural dataflow can be summarized as:

reviews.csv (HDFS)

|

↓

Remove header

|

↓

Remove null ratings

|

↓

Convert rating to float

|

↓

Filter rating >= 1.0

|

↓

Group by product category

|

↓

Calculate AVG(rating)

|

↓

DUMP final averages

7. Conclusion

The Apache Pig program successfully processed the e-commerce review dataset stored in HDFS. Invalid ratings below 1.0 and null ratings were removed, the remaining reviews were grouped by product category, and the average rating for each category was calculated successfully.